

AI-Powered Enterprise E-Commerce Platform

User Stories Document

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1. Introduction

User stories describe system requirements from the user's perspective.

They focus on user needs rather than technical implementation details.

User stories help:

- understand user expectations,
 - prioritize features,
 - improve communication,
 - support Agile development.
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2. Purpose

The purpose of this document is to identify all major interactions between users and the system.

This document will later support:

- Use Case Diagrams
 - Sprint Planning
 - Product Backlog Creation
 - System Design
 - Testing Activities
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3. User Roles

The following users have been identified.

1. Customer
 2. Administrator
 3. Inventory Manager
 4. Business Owner
 5. Delivery Personnel
 6. AI Services
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4. User Story Template

All user stories follow the format:

```
As a <user>,  
I want <goal>,  
So that <benefit>.
```

5. Customer User Stories

US-001 User Registration

As a customer,

I want to create an account,

So that I can purchase products and manage my orders.

Priority: High

US-002 User Login

As a customer,

I want to login securely,

So that my personal information remains protected.

Priority: High

US-003 View Profile

As a customer,

I want to view and update my profile,

So that my information remains accurate.

Priority: Medium

US-004 Search Products

As a customer,

I want to search products,

So that I can quickly find products that I need.

Priority: High

US-005 Filter Products

As a customer,

I want to filter products,

So that I can narrow down my search results.

Priority: Medium

US-006 View Product Details

As a customer,

I want to view detailed product information,

So that I can make informed purchasing decisions.

Priority: High

US-007 Add Products to Cart

As a customer,

I want to add products to my shopping cart,

So that I can purchase multiple products together.

Priority: High

US-008 Manage Shopping Cart

As a customer,

I want to modify product quantities,

So that I can control my purchases.

Priority: High

US-009 Wishlist Management

As a customer,

I want to save products to my wishlist,

So that I can purchase them later.

Priority: Medium

US-010 Place Orders

As a customer,

I want to place orders,

So that I can purchase products online.

Priority: Critical

US-011 View Order History

As a customer,

I want to view previous orders,

So that I can track my purchases.

Priority: Medium

US-012 Track Orders

As a customer,

I want to track delivery status,

So that I know when my products will arrive.

Priority: High

US-013 Product Reviews

As a customer,

I want to provide ratings and reviews,

So that I can share my experiences.

Priority: Medium

US-014 Receive Recommendations

As a customer,

I want personalized recommendations,

So that I can discover products that match my interests.

Priority: High

US-015 Chat with AI Assistant

As a customer,

I want an AI assistant,

So that I can receive help quickly.

Priority: Medium

6. Administrator User Stories

US-016 Manage Users

As an administrator,

I want to manage customer accounts,

So that I can maintain platform security.

Priority: High

US-017 Manage Products

As an administrator,

I want to add, edit, and remove products,

So that product information remains accurate.

Priority: Critical

US-018 Manage Categories

As an administrator,

I want to manage categories,

So that products remain organized.

Priority: High

US-019 Monitor Transactions

As an administrator,

I want to monitor transactions,

So that suspicious activities can be identified.

Priority: Critical

US-020 View Analytics

As an administrator,

I want analytics dashboards,

So that I can understand business performance.

Priority: High

US-021 Manage Orders

As an administrator,

I want to update order status,

So that customers receive accurate information.

Priority: High

US-022 Receive Fraud Alerts

As an administrator,

I want fraud notifications,

So that I can investigate suspicious activities.

Priority: Critical

US-023 Monitor AI Services

As an administrator,

I want to monitor AI services,

So that system performance remains stable.

Priority: Medium

7. Inventory Manager User Stories

US-024 Monitor Stock Levels

As an inventory manager,

I want to monitor inventory,

So that stock shortages can be prevented.

Priority: High

US-025 Update Inventory

As an inventory manager,
I want to update stock quantities,
So that inventory information remains accurate.
Priority: High

US-026 Receive Low Stock Alerts

As an inventory manager,
I want low stock notifications,
So that products can be reordered.
Priority: High

US-027 View Forecast Reports

As an inventory manager,
I want demand forecasting reports,
So that inventory planning becomes easier.
Priority: High

8. Business Owner User Stories

US-028 View Revenue Reports

As a business owner,
I want revenue reports,
So that I can evaluate business performance.
Priority: High

US-029 View Customer Analytics

As a business owner,
I want customer insights,
So that I can make strategic decisions.

Priority: High

US-030 View Product Performance

As a business owner,

I want product analytics,

So that I can identify high-performing products.

Priority: High

US-031 View Forecasting Reports

As a business owner,

I want future sales predictions,

So that I can plan business activities.

Priority: High

US-032 Ask Business Questions

As a business owner,

I want an AI assistant,

So that I can obtain intelligent insights.

Examples:

- Which products generated the highest revenue?
- Which customers generate the most profit?
- What are expected sales next month?

Priority: Medium

9. Delivery Personnel User Stories

US-033 View Delivery Information

As delivery personnel,

I want delivery information,

So that products can be delivered accurately.

Priority: Medium

US-034 Update Delivery Status

As delivery personnel,

I want to update delivery status,

So that customers receive accurate updates.

Priority: Medium

10. AI Service User Stories

US-035 Recommendation System

As an AI recommendation service,

I want customer data,

So that I can generate personalized recommendations.

US-036 Fraud Detection System

As a fraud detection service,

I want transaction information,

So that suspicious activities can be detected.

US-037 Forecasting Service

As a forecasting service,

I want historical sales data,

So that future demand can be predicted.

US-038 Customer Segmentation Service

As an AI service,

I want customer behavior information,

So that customers can be grouped effectively.

US-039 AI Chat Assistant

As an AI assistant,

I want business and customer information,

So that intelligent responses can be generated.

11. Epic User Stories

Large functionalities can be grouped into Epics.

Epic 1

Customer Management

Includes:

US-001 to US-003

Epic 2

Product Management

Includes:

US-004 to US-009

Epic 3

Order Management

Includes:

US-010 to US-012

Epic 4

Administration

Includes:

US-016 to US-023

Epic 5

AI Services

Includes:

US-035 to US-039

12. Acceptance Criteria

Examples:

US-001 Acceptance Criteria

- User can successfully register.
 - Duplicate emails are rejected.
 - Password validation is performed.
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US-010 Acceptance Criteria

- User can place orders successfully.
 - Inventory updates automatically.
 - Confirmation notification is generated.
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US-022 Acceptance Criteria

- Suspicious activities generate alerts.
 - Administrators receive notifications.
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13. Story Prioritization

MoSCoW Prioritization Technique:

Must Have

- Registration
 - Login
 - Product Search
 - Cart
 - Orders
 - Product Management
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Should Have

- Recommendations
- Forecasting

- Notifications

Could Have

- Wishlist
- AI Chat Assistant

Won't Have (Initial Version)

- Mobile Application
- International Shipping
- Multi-vendor Marketplace

14. Story Traceability Matrix

Business Goal	User Stories
Improve customer satisfaction	US-004, US-014
Increase sales	US-014, US-031
Reduce fraud	US-022, US-036
Improve inventory management	US-027, US-037
Improve decision making	US-020, US-032

15. Product Backlog (Initial)

Story ID	Description	Priority
US-001	Registration	High
US-002	Login	High
US-004	Product Search	High
US-007	Add to Cart	High
US-010	Place Orders	Critical
US-017	Product Management	Critical
US-022	Fraud Alerts	Critical
US-031	Forecast Reports	High

16. Conclusion

This document defines the user expectations and interactions of the AI-Powered Enterprise E-Commerce Platform.

These user stories provide the foundation for:

- Use Case Diagrams
- Sprint Planning
- Product Backlogs
- System Design
- Testing Activities
- Agile Development Processes

References

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